Eigenvalues and inverse problems

A chromatic lower bound for positive-semidefinite nullity with SAP

Difficulty: challenging **Importance:** interesting to specialist

Status: Lean verified

Rating rationale (historical): Challenging reflects the need for a stronger general PSD/SAP realization bound than current connectivity estimates supply; implication from Hadwiger does not establish equivalent difficulty. Specialist impact reflects the focus on this particular graph nullity invariant.

Lean proof and verification evidence — 2026-10-18

The self-contained Lean project proves *NLA.SP12.target*, the complete original PSD/SAP/chromatic assertion. Independent exact statement and source reviews, fresh kernel audits and the actual isolated Linux Comparator/LeanCert kernel receipt pass with only the three standard foundational axioms. The checked proof commit is *529e7b0560cef6461c98e950046860a160e6f86c*, run 39142969327. The original page, TeX and PDF are preserved byte for byte. Mathematical attribution and all original text and history below remain unchanged. Formalization and independent reviews are AI-agent work, not external human peer review.

Literature-dependent resolution - 2026-09-11

Solved. Hall's **Corollary 3.22**, based on **Theorem 3.20**, gives $\nu(H) \geq \delta(H)$ for every graph H. Induced-subgraph monotonicity and a vertex-critical subgraph of chromatic number $\chi(G)$ then prove the exact target $\nu(G) \geq \chi(G) - 1$. The application note supplies the complete monotonicity and coloring argument, including rank-zero and disconnected cases.

The all-graph theorem is due to **H. Tracy Hall**, *The Delta Theorem*, arXiv:2601.01211v1, submitted 3 January 2026. **George Stepaniants**, Department of Computing and Mathematical Sciences, California Institute of Technology, is the author of the explanatory application note; no new theorem discovery or priority is claimed. Application PDF · Standalone TeX.

A separate Codex-agent review checked the full essential proof in Hall's preprint and the exact deduction and returned **PASS**. The source remains a preprint; this is independent automated-agent review, not external human peer review or formal verification. Substantial ChatGPT/Codex assistance and the review's precise limits are disclosed in the submission record.

The earlier status checks below are retained as historical records. The original ID, target, path, historical ratings and attributed partial results remain unchanged. Unsupported prior artifact and graph-atlas claims remain withdrawn.

Problem statement

Let G be a finite simple undirected graph on $n \geq 1$ vertices. Let $\mathcal{S}(G)$ be the real symmetric matrices whose off-diagonal nonzero entries occur exactly at edges of G, with unrestricted diagonal.

A matrix $A \in \mathcal{S}(G)$ has the **strong Arnold property (SAP)** if the only real symmetric matrix X satisfying

$$AX = 0, \qquad A \circ X = 0, \qquad I_n \circ X = 0$$

is $X = 0$, where $\circ$ is the entrywise product. Define

$$\nu(G) = \max\{\dim \ker A : A \in \mathcal{S}(G),\ A \succeq 0,\ A \text{ has SAP}\}.$$

Writing $\chi(G)$ for the minimum number of colors in a proper vertex coloring, does every G satisfy

$$\nu(G) \geq \chi(G) - 1?$$

Relevance and ratings

The conclusion asserts the existence of a positive-semidefinite matrix, with an exact prescribed sparsity pattern, of rank at most $n - \chi(G) + 1$ and with an additional nondegeneracy property. Equivalently it constrains structured Gram matrix construction. This is adjacent to core NLA through matrix realization and inverse eigenvalue theory.

References

- F. Barioli, S. M. Fallat, H. Gupta, and Z. Li, *The weak version of the graph complement conjecture and partial results for the delta conjecture*, Discrete Mathematics 349 (2026), 114861, Conjecture 4.1 and §1 definitions (primary manuscript; journal).
- F. Barioli, S. M. Fallat, and L. Hogben, *A variant on the graph parameters of Colin de Verdière: Implications to the minimum rank of graphs*, Electronic Journal of Linear Algebra 13 (2005), 387–404, pp.388–390 on SAP and nullity parameters (primary paper).
- M. Chudnovsky and A. Ovetsky Fradkin, *Hadwiger's conjecture for quasi-line graphs*, Journal of Graph Theory 59 (2008), 17–33, Theorem 1.1 (primary manuscript; journal).

Status check

This exact inequality is expressly proposed in the 2026 paper. Its weaker independence-number consequence, Conjecture 4.4, is not separately counted. The source explains that the strong delta conjecture and Hadwiger's conjecture would each imply the assertion; the original unrestricted symmetric delta conjecture in the companion entry is a different statement. Searches for this paper, "nu", "chromatic", "positive semidefinite", "conjecture", and 2026 found no resolution. No claim is made that these conjectures are logically independent.

The 2026 counterexamples to a Colin de Verdière conjecture about Laplace eigenvalue multiplicities on surfaces concern a different invariant; they do not refute this finite PSD/SAP inequality.

Audit update — 2026-09-10

Partially resolved: the inequality holds for quasi-line graphs, meaning each vertex neighborhood is the union of two cliques. This follows by combining Chudnovsky–Fradkin Theorem 1.1 with the proved inequality $\nu(G) \geq \eta(G) - 1$ recalled immediately before Conjecture 4.1, where η is the largest clique-minor order. This is an implication of two established results, not a claim that the general conjecture is proved. Both full primary texts, the 2026 publication record, and targeted later searches were checked; no general resolution was located.